Module Overview

Essential Question: What do plants need to live and grow?
Quote: "The secret garden bloomed and bloomed and every morning revealed new miracles." — Frances Hodgson Burnett
Big Idea Words: fertilize, germinate, survive
Main Stories (workbook 4):

- Experiment with What a Plant Needs to Grow · Nadia Higgins · Informational Text
- Jack and the Beanstalk · Helen Lester (illus. Jesús Aguado) · Fairy Tale
- Jackie and the Beanstalk · Lori Mortensen (illus. Ben Scruton) · Fairy Tale
- Don't Touch Me! · Elizabeth Preston · Informational Text

Note. "The Growth of a Sunflower" (pp.100–101) is a Short Read photo essay — excluded from workbook main 4 per convention. "George Washington Carver: The Wizard of Tuskegee" (pp.154–157) is a StoryBots video selection, not a print main read — excluded. Mr. Fefifofum in "Jackie and the Beanstalk" is an upstairs human tenant who is "as big and loud as a giant" (simile), NOT a literal giant.

Answer Distribution (79 Multiple-Choice Items)

Verified counts across Worksheets A (27), B (32), and D (20) combined. Worksheet C is open-response IB and not counted.

A · 19 · B · 20 · C · 21 · D · 19

Balance enforced via choice-order rotation only — correct-answer meanings and option texts are preserved from the Source of Truth. All 27 Big Idea + Power Words covered in mybook page-order on Worksheet A.

Voca Trap Items (Worksheet A)

8 of 27 ≈ 30% — meets G2 trap target band.

- Q5 **process** — polysemy (parade / line of people vs. set of steps)
- Q9 **moisten** — sound-trap with "moan" (moisten ≠ moan)
- Q11 **spiky** — distractor "popular drink" pulls casual usage
- Q12 **swipe** — polysemy (credit-card payment vs. quick grab)
- Q15 **plenty** — homophone-near trap with "plain" (flat field) + "small amount"
- Q19 **glanced** — polysemy (look quickly vs. "glanced off" = hit at angle)
- Q23 **extra** — polysemy ("an extra" in a movie vs. additional)
- Q25 **nasty** — sound-trap with "knotty" (shaped like a knot)
- Q27 **attack** — sound-trap with "attach" (stick onto a page)

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Bean in the Cup · **120 words** · Lexile 240–290L · RIT 180–189
- Passage 2 · The Cactus and the Boy · **132 words** · Lexile 300–350L · RIT 190–195
- Passage 3 · The Tomato That Almost Died · **149 words** · Lexile 360–410L · RIT 195–199
- Passage 4 · The Forest That Cleans the Air · **150 words** · Lexile 440–500L · RIT 199–200

All four passages within target band 80–150 words; RIT 180→200; Lexile 240L–500L. Each passage models a different facet of the EQ — a child growing a single bean (germinate / seedling), a cactus surviving a dry place (defense + storage), a near-death recovery requiring daily care (fertilize / provide), and the forest as a planet-scale system (process / oxygen / minerals).

Worksheet A — Vocabulary · Answer Key

27 items · CCSS L.2.4 / L.2.5 / L.2.6

BIG IDEA WORDS

Q1 L.2.6

fertilize

D

Meaning: add something to soil that helps plants grow. **Source:** SoT Big Idea Word 8.1 — OCR lines 136, 164, 166.

Q2 L.2.6

germinate

C

Meaning: begin to grow. **Source:** SoT Big Idea Word 8.2 — OCR lines 136, 177, 179.

Q3 L.2.6

survive

A

Meaning: stay alive. **Source:** SoT Big Idea Word 8.3 — OCR lines 136, 170, 171.

POWER WORDS · EXPERIMENT WITH WHAT A PLANT NEEDS TO GROW

Q4 L.2.4

minerals

B

Evidence: "They need minerals to stay healthy" (SoT, OCR 605). Small pieces from the earth.

Q5 L.2.5 TRAP

process

D

Evidence: "photosynthesis... a process in which green leaves make food" (SoT, OCR 419). **Trap:** B "parade" — polysemy.

Q6 L.2.4

fuels

A

Evidence: "Sunlight fuels photosynthesis" (SoT, OCR 419). Gives the energy that makes something work.

Q7 L.2.4

provides

C

Source: SoT Power Words sidebar pg-05; "Most plants get minerals from the soil" (OCR 846). Gives what is needed.

Q8 L.2.4

sprout

B

Evidence: "Water softens the coat, so the seed can sprout" (SoT, OCR 591). Begin to grow out of a seed.

Q9 L.2.5 TRAP

moisten

A

Evidence: "Moisten the second bowl's cotton balls" (SoT, OCR 733). **Trap:** D "moan" — sound-trap.

Q10 L.2.4

seedling

D

Evidence: "Give the seedling water and sun" (SoT, OCR 296). A young plant just beginning to grow.

Q11 L.2.5 TRAP

spiky

C

Source: SoT Power Words sidebar pg-05. **Trap:** B "popular drink" — distractor.

POWER WORDS · JACK AND THE BEANSTALK

Q12 L.2.5 TRAP

swipe

B

Evidence: "SWIPE!—down the beanstalk he went" (SoT, OCR 1328). Quick grab/sweep. **Trap:** C credit-card swipe — polysemy.

Q13 L.2.4

whacked

A

Evidence: "Jack grabbed an ax, whacked the beanstalk" (SoT, OCR 1475). Hit hard.

Q14 L.2.4

whimpered

D

Source: SoT Power Words sidebar pg-12. A low, soft, sad cry.

Q15 L.2.5 TRAP

plenty

B

Evidence: "had plenty to eat... happily ever after" (SoT, OCR 1546). **Trap:** D "plain" — sound-near + A "small amount" — opposite-meaning trap.

POWER WORDS · JACKIE AND THE BEANSTALK

Q16 L.2.4

adorable

C

Evidence: "cute little calf" (SoT, OCR 1779, 1859). Very cute and lovable.

Q17 L.2.4

oversized

A

Evidence: "they lived with a GIGANTIC cow" (SoT, OCR 2571). Much bigger than usual.

Q18 L.2.4

hauling

D

Source: SoT Power Words sidebar pg-19. Pull or carry something heavy.

Q19 L.2.5 TRAP

glanced

B

Source: SoT Power Words sidebar pg-19. Took a quick look. **Trap:** A "glanced off" — polysemy.

POWER WORDS · DON'T TOUCH ME!

Q20 L.2.4

sharp

C

Evidence: "Roses have sharp prickles on their stems" (SoT, OCR 2598). Comes to a fine point that can cut or poke.

Q21 L.2.4

prickles

A

Evidence: "sharp prickles on their stems" (SoT, OCR 2598). Small sharp points on a plant.

Q22 L.2.4

thorns

D

Evidence: "Acacia trees have big, scary thorns" (SoT, OCR 2806). Sharp woody points from a stem or branch.

Q23 L.2.5 TRAP

extra

B

Evidence: "for extra protection, they use ants" (SoT, OCR 2807). **Trap:** A "an extra in a movie" — polysemy.

Q24 L.2.4

poke

A

Source: SoT Power Words sidebar pg-26. Press or jab with a sharp point.

Q25 L.2.5 TRAP

nasty

C

Evidence: "You might get a nasty rash" (SoT, OCR 3708). **Trap:** B "knotty" — sound-near trap.

Q26 L.2.4

sensitive

B

Evidence: "the sensitive or touch-me-not plant... folds up its leaves" (SoT, OCR 2702). Quick to react to touch.

Q27 L.2.5 TRAP

attack

C

Evidence: "the ants attack and sting" (SoT, OCR 2819). **Trap:** A "attach" — sound-near trap.

Worksheet B — Reading Comprehension · Answer Key

32 items · CCSS RL.2.1–2.7 / RI.2.1–2.8

Story 1 · Experiment with What a Plant Needs to Grow

INFORMATIONAL · 8 ITEMS

Q1 RI.2.1

B

Evidence: "Plants need air and water. They need minerals to stay healthy." + sunlight fuels photosynthesis (SoT, OCR 605, 608).

Q2 RI.2.8

C

Reason→Point: The author tells the reader to photograph the plant on day one so you can "compare the way the plant looks at the end to how it looked at the beginning and see how much it has changed" (SoT key line: "Compare what you see to the photo you took on the first day"; TP OCR 714–717, 760–761).

Q3 RI.2.3

C

Evidence: "1. Start by putting two cotton balls in each of the bowls" (SoT, OCR 444).

Q4 RI.2.3

D

Evidence: "A seed has a coat that protects it. Water softens the coat, so the seed can sprout" (SoT, OCR 591).

Q5 RI.2.3

A

Evidence: "Petroleum jelly kept some of the leaves from letting air in and out. Those leaves started to wilt" (SoT, OCR 650).

Q6 RI.2.5

B

Source: SoT genre — Informational Text. Real facts + steps.

Q7 RI.2.6

D

Author's Purpose: The text explains what plants need and walks readers through hands-on experiments — the author's purpose is to teach AND show how to find out (SoT key lines: "Plants need air and water...", "Steps: 1. Start by putting two cotton balls...", "Measuring helps scientists show exactly what is happening").

Q8 RI.2.2

C

Synthesis: Air + water + sunlight + minerals = needs; experiments prove it (SoT, OCR 359, 419, 591). **Trap:** A overstates sunlight (photosynthesis ≠ only need); B overstates soil; D confuses with Story 1 (Sunflower photo essay).

Story 2 · Jack and the Beanstalk

FAIRY TALE · 8 ITEMS

Q9 RL.2.1

B

Evidence: "a boy named Jack lived with his mother in a poor little hut" (SoT, OCR 1022).

Q10 RL.2.3

C

Evidence: "Jack returned alone with a handful of beans... like a treasure" (SoT, OCR 1144).

Q11 RL.2.3

A

Evidence: "planted the pretty beans in a sunny spot. He gave them a drink of water" (SoT, OCR 1180).

Q12 RL.2.3

D

Evidence: "SWIPE!—down the beanstalk he went with his treasure" (SoT, OCR 1328).

Q13 RL.2.3

A

Evidence: "Jack grabbed an ax, whacked the beanstalk, and SPLAT" (SoT, OCR 1475).

Q14 RL.2.3

B

Evidence: "First, he showed the giant how to water the beanstalk. Together, they cared for the plant" (SoT, OCR 1521).

Q15 RL.2.5

C

Source: SoT genre — Fairy Tale. "Once upon a time" + magical beanstalk overnight (SoT, OCR 1018, 1238).

Q16 RL.2.2

D

Synthesis: Seed + care + helping the giant → plenty for everyone (SoT, OCR 1521, 1546). **Trap:** A contradicts OCR 1521 (Jack teaches the giant); B is only the start of the story; C is surface/literal, not the theme.

Story 3 · Jackie and the Beanstalk

FAIRY TALE · 8 ITEMS

Q17 RL.2.1

B

Evidence: "Jackie lived in an apartment with her father... they lived with a GIGANTIC cow" (SoT, OCR 1855, 1858).

Q18 RL.2.3

A

Evidence: "Buying the cow had been a stupendously bad idea, but nobody thought of that when it was a cute little calf" (SoT, OCR 1859).

Q19 RL.2.3

C

Evidence: "The woman had no money, but she did have some special beans" (SoT, OCR 1938).

Q20 RL.2.3

D

Evidence: "Plant them in dirt, add water, and wait. By morning, SHAZAM!" (SoT, OCR 1959, 1966).

Q21 RL.2.3

B

Evidence: "It belonged to Mr. Fefifofum, an upstairs tenant who was as big and loud as a giant" (SoT, OCR 2056). **Caution:** simile only — NOT a literal giant (SoT hallucination warning).

Q22 RL.2.3

A

Evidence: "Jackie gave it back, apologized, and explained" (SoT, OCR 2199).

Q23 RL.2.9

D

Compare/Contrast: Setting (hut → apartment); giant (real → simile-neighbor); ending (keep → return harp). SoT page-ranges 116–129 vs 130–143.

Q24 RL.2.2

C

Synthesis: Modern Jackie + return + apology + explain (SoT, OCR 2199). Beanstalk works in any setting; the right thing to do is constant. **Trap:** A contradicts OCR 2199; B is the SoT hallucination warning (Mr. Fefifofum is a simile, not a real giant); D contradicts OCR 1859 ("stupendously bad idea").

Story 4 · Don't Touch Me!

INFORMATIONAL · 8 ITEMS

Q25 RI.2.2

A

Evidence: "Plants can't run away from hungry animals. So some of them fight back. They have defenses" (SoT, OCR 2551).

Q26 RI.2.3

C

Evidence: "Roses have sharp prickles on their stems" (SoT, OCR 2598).

Q27 RI.2.3

B

Evidence: "Holly leaves and berries have extra protection. They're poisonous!" (SoT, OCR 2604).

Q28 RI.2.5

D

Text Structure: The author moves plant by plant — roses, holly, cactus, poison ivy, stinging nettle, sensitive plant, acacia — naming each plant and the defense it uses. The structure is example-by-example (SoT key lines show this sequence).

Q29 RI.2.3

A

Evidence: "If you touch it, the plant quickly folds up its leaves" (SoT, OCR 2702).

Q30 RI.2.3

B

Evidence: "for extra protection, they use ants. If a hungry animal tries to take a bite, the ants attack and sting it" (SoT, OCR 2807, 2819).

Q31 RI.2.6

C

Author's Purpose: Teach readers how plants protect themselves to survive (SoT genre — Informational; survive is a Big Idea word). **Trap:** A overgeneralizes (not every plant is poisonous); B treats it as a how-to-grow guide (it isn't); D narrows to one plant (text covers many).

Q32 RI.2.2

D

Synthesis: No running, many defenses — prickles, thorns, poisons, sensitive folding, ant helpers — together = survival. **Trap:** A directly contradicts OCR 2551 ("Plants can't run away"); B narrows to two examples; C makes poison ivy the focus (it's one example among many).

Worksheet C — IB Critical Thinking · Answer Key

5 questions · PYP Key Concepts · DOK 2–4

Q1 Function DOK 3

In **"Experiment with What a Plant Needs to Grow,"** show what happens when one of the four needs (air, water, sunlight, minerals) is missing — and what happens when all four are there.

MODEL ANSWER

Missing one need: When the leaves were coated in petroleum jelly, they could not let air in and out — they started to WILT. In the seed bowl with no water, the seed never softened — it stayed dry and never sprouted. Take away even one need and the plant starts to die. **All four needs are there:** Sunlight FUELS photosynthesis. The leaves take in air and water. The soil PROVIDES minerals. The seed SPROUTS, grows into a SEEDLING, and finally blooms. **The easiest to forget:** Air. We see the plant get water and sun, but we rarely see it breathe. Yet petroleum jelly on a leaf — blocking air — was enough to make leaves wilt. Air is invisible, but the plant needs it the same way it needs water.

KEYWORDS (3)

petroleum jelly → wilt | seed coat softens → sprout | air is the easiest to forget

SCORING NOTES

- Full credit (3): one missing need + result, all four needs working together + result, and a reflection naming the easiest need to forget with a reason.
- Partial (2): missing-need described + all-four described, but no reflection.
- Minimal (1): only "plants need water" without text evidence.

TEACHING STEPS

Step 1 · Name the four needs.

강사: *"Without looking, name the four things a plant needs. Use your fingers — air, water, sunlight, minerals."*

코멘트. 손가락 네 개로 네 가지 필요 — 공기, 물, 햇빛, 미네랄.

Step 2 · Cover one finger.

강사: *"Now cover ONE finger — that need is gone. What happens to the plant? Use evidence from the experiment text."*

코멘트. 손가락 하나 가리기 — 그 필요가 사라지면? 본문 실험 근거 사용.

Step 3 · The invisible one.

강사: *"Which need is invisible? Why is air the one most often forgotten?"*

코멘트. 보이지 않는 필요는? 왜 공기가 제일 잊혀지는가?

STUDENT PROMPTS

- "What did the petroleum jelly do to the leaves?"
  바셀린은 잎에게 무엇을 했나?
- "Which need is the soil giving the plant?"

흙은 식물에게 어떤 필요를 주는가?

- "If you had to give up ONE need, which would the plant survive longest without?"
  하나만 빼면, 식물이 가장 오래 견딜 수 있는 필요는?

DIFFERENTIATION

STRUGGLING Provide 4 picture cards (sun, water, air, soil) and ask: "Which one was missing in the petroleum-jelly experiment?"

4장 그림 카드 + 한 가지 빠진 실험 매칭.

ADVANCED Design a fifth experiment that would test the importance of MINERALS specifically.

미네랄만 따로 검증할 5번째 실험 설계.

COMMON MISCONCEPTIONS

PITFALL "Plants make food from soil." Plants make food from sunlight + air + water through photosynthesis. Soil provides minerals (a need), not food.

식물은 흙을 먹지 않음 — 광합성으로 햇빛·공기·물로 음식 만듦. 흙은 미네랄(필요) 공급원.

PITFALL "Water is the only need." All four needs work together. Take away ONE and the plant fails.

물만 필요한 게 아님. 네 가지가 함께 — 하나만 빠져도 안 됨.

LEARNER PROFILE LINK

INQUIRERS The text asks "Can seeds get too much water?" — that question-first stance is the inquirer's habit.

질문으로 시작하는 자세 — Inquirer.

KOREAN BRIDGE

표현 "It needs four things to live." → "이건 살기 위해 네 가지가 필요해."

"네 가지 필요"는 G2 과학 핵심 문장 — 한국어로도 그대로 사용 가능.

개념 "photosynthesis / 광합성 — minerals / 미네랄 — sprout / 싹트다"

핵심 단어 영-한 대응.

Q2 Change DOK 2

In **"Experiment with What a Plant Needs to Grow"** and "The Growth of a Sunflower," a seed changes into a blooming plant in steps. Make a 3-step life cycle.

MODEL ANSWER

Step 1 · Seed · Germinate: A seed has a coat. Water softens the coat. The seed begins to grow — this is GERMINATION. A tiny green tip pushes up. **Step 2 · Seedling · Grow:** The young plant is now a SEEDLING. It still needs water and sun. The leaves take in light and air. The roots pull minerals from the soil. The seedling grows taller and stronger each day. **Step 3 · Bloom:** Soon a bud develops. The plant fertilizes itself with sunlight + water + minerals. BLOOM! The plant makes flowers (and more seeds). The cycle is ready to start again. **The hardest step:** Step 1 — germinate. Underground, the seed has to break its own coat. We can't see it. It's the moment with the highest chance of failure. Once it has its first leaf, the rest is care.

KEYWORDS (3)

seed coat → sprout | seedling needs water + sun | bud → BLOOM

SCORING NOTES

- Full credit (3): 3 named steps in order + 1 reflection on the hardest step with reason.
- Partial (2): 3 steps but order or names unclear, OR no reflection.
- Minimal (1): "the seed grows" without naming the stages.

TEACHING STEPS

Step 1 · Draw the cycle.

강사: *"Draw 3 boxes left to right — seed, seedling, bloom. Inside each, draw what the plant looks like."*

코멘트. 3박스 — 씨앗·새싹·꽃, 각 박스에 그림 그리기.

Step 2 · Name what changes.

강사: *"Between each box, write ONE thing that changes. Use 'germinate,' 'grow,' 'bloom.'"*

코멘트. 각 박스 사이에 변화 한 가지 — germinate / grow / bloom.

Step 3 · Find the invisible step.

강사: *"Which step happens UNDER the dirt where we can't see? Star it."*

코멘트. 흙 아래에서 일어나는 단계는? 별표.

STUDENT PROMPTS

- "What does a seed need to GERMINATE?"
  씨앗이 싹트려면 무엇이 필요한가?
- "How is a seedling different from a seed?"
  새싹은 씨앗과 어떻게 다른가?
- "Which step would a careless gardener miss the most?"
  부주의한 정원사가 가장 놓치는 단계는?

DIFFERENTIATION

STRUGGLING Pre-cut 3 picture cards (seed, seedling, flower). Student puts them in order.

3장 그림 카드 순서 맞추기.

ADVANCED Add a 4th step — what happens AFTER bloom? (Seeds, new cycle.) Connect to "Forest That Cleans the Air" passage.

4단계 추가 — bloom 다음은? 새 씨앗·새 cycle.

COMMON MISCONCEPTIONS

PITFALL "Seeds wake up by themselves." Seeds need water + warmth — without those they stay dormant.

씨앗은 저절로 깨어나지 않음. 물과 온도가 필요.

PITFALL "Bloom is the end." Bloom is also the START of new seeds — the cycle continues.

꽃은 끝이 아님. 새 씨앗의 시작. 순환은 계속됨.

LEARNER PROFILE LINK

KNOWLEDGEABLE Knowing the life cycle = knowing how living things work — Knowledgeable.

생애주기를 안다는 것 — Knowledgeable.

KOREAN BRIDGE

표현 "First the seed germinates, then the seedling grows, then it blooms." → "먼저 씨앗이 싹트고, 새싹이 자라고, 꽃이 핀다."

3단계 한 문장 골격 — 글쓰기 그대로 사용.

Q3 Connection DOK 3

Compare **"Jack and the Beanstalk"** and **"Jackie and the Beanstalk."** Both use a beanstalk — but the two stories make different choices.

MODEL ANSWER

Jack · Setting / Helper: A poor little hut in the countryside. Jack's helper is his own mother (after she gets over her anger). The cow Moozetta starts the story. **Jackie · Setting / Helper:** A crowded apartment in the city. Jackie's helper is a mysterious woman with special beans. The GIGANTIC cow lives with them in the apartment. **Jack · What he KEEPS:** He swipes the giant's harp and goose, brings them down the beanstalk, and KEEPS them. (Later he also helps the giant grow his own beanstalk.) **Jackie · What she RETURNS:** She also borrows the harp, but realizes she did not ASK. She returns it, apologizes, and explains. **Which feels more right:** Jackie's choice. Both stories show a small bean growing into something powerful — but Jackie shows that doing the RIGHT thing means asking, returning, and apologizing. Jack's old story keeps; Jackie's new story returns.

KEYWORDS (3)

hut vs. apartment | real giant vs. as-big-as-a-giant neighbor | keep vs. return

SCORING NOTES

- Full credit (3): 4 web-cell entries filled accurately + a reasoned choice for which feels more right.
- Partial (2): 4 cells but choice not reasoned.
- Minimal (1): only "the stories are different" — no specifics.

TEACHING STEPS

Step 1 · Two columns.

강사: *"Draw two columns — Jack / Jackie. List one detail per row."*

코멘트. 2열 — Jack / Jackie, 한 줄에 하나씩.

Step 2 · The harp moment.

강사: *"Find the page where each story handles the harp. What does each character DO?"*

코멘트. 하프 장면 — 두 인물이 각각 무엇을 했나?

Step 3 · Modern vs. classic.

강사: *"Which character would your parents tell you to act like? Why?"*

코멘트. 부모님이라면 어느 쪽처럼 행동하라고 할까? 이유는?

STUDENT PROMPTS

- "Why is Mr. Fefifofum NOT a real giant?"
  왜 페피포펌 씨는 진짜 거인이 아닌가?
- "In Jack's story, was it OK to take the harp? Was it OK in Jackie's?"
  잭의 이야기에서 하프를 가져온 것은 괜찮은가? 재키의 이야기에서는?
- "What does each story say about RULES?"
  각 이야기는 규칙에 대해 무엇을 말하는가?

DIFFERENTIATION

STRUGGLING Provide a 4-cell Venn-style grid pre-labeled (setting / helper / kept / returned). Student fills 1 word per cell.

4셀 격자 — 단어 1개씩.

ADVANCED Write a third version: "Yujin and the Beanstalk" — set in your own town, with your own moral choice.

자기 동네 버전 'Yujin and the Beanstalk' 한 문단 쓰기.

COMMON MISCONCEPTIONS

PITFALL "Mr. Fefifofum is a real giant." The text uses simile — "AS big and loud AS a giant." He is a tall, loud upstairs neighbor.

페피포펌은 진짜 거인이 아니라 비유. '거인처럼 크고 시끄러운' 윗집 이웃.

PITFALL "Jackie just copies Jack." Jackie's version retells the SAME plot but makes DIFFERENT moral choices — that is what makes it a new story.

재키는 그냥 모방이 아님. 같은 이야기 + 다른 도덕적 선택.

LEARNER PROFILE LINK

PRINCIPLED Jackie returns the harp because it is the right thing to do — Principled.

재키가 하프를 돌려준 행동 — Principled.

KOREAN BRIDGE

표현 "I borrowed it without asking. I am sorry. Here it is back." → "묻지 않고 빌렸어요. 죄송합니다. 돌려드릴게요."

사과·반환 3문장 한 묶음 — 학생 일상 적용 가능.

Q4 Form DOK 3

In **"Don't Touch Me!,"** plants have three KINDS of defense. Trace one example of each — and how it helps the plant survive.

MODEL ANSWER

Sharp · Prickles or Thorns: The ROSE has sharp prickles on its stem. The HOLLY leaf has very sharp points. The ACACIA tree has big, scary thorns. If a hungry animal tries to bite, the sharp points POKE — and the animal stops. The plant survives because biting hurts. **Chemical · Poisons or Itch:** HOLLY leaves and berries are POISONOUS — extra protection. POISON IVY has oils that make you itchy. The chemical makes the animal sick OR uncomfortable. Once an animal learns, it does not come back. **Helper · Folding or Ants:** The SENSITIVE plant (touch-me-not) does not stab. When touched, it quickly FOLDS up its leaves — so the animal sees no food. The ACACIA tree uses ANTS as helpers. If a hungry animal tries to take a bite, the ants ATTACK and sting it. The plant doesn't fight — the ants do. **Most surprising:** The ant helper. A tree using ants as a tiny army is more like a partnership than a defense.

KEYWORDS (3)

rose / holly / acacia → sharp | holly / poison ivy → chemical | sensitive plant / acacia + ants → helper

SCORING NOTES

- Full credit (3): 1 example for each of 3 defense kinds + how it helps + a most-surprising reflection.
- Partial (2): 3 examples but no reflection.
- Minimal (1): only "plants have thorns" — single kind only.

TEACHING STEPS

Step 1 · Three columns.

강사: *"Draw 3 columns — Sharp / Chemical / Helper. List one plant in each."*

코멘트. 3열 — 날카로움 / 화학 / 도우미, 각각 식물 한 개.

Step 2 · How does it stop the animal?

강사: *"For each plant, write ONE sentence: how does this defense stop a hungry animal?"*

코멘트. 각 식물 — 어떻게 배고픈 동물을 막는가? 한 문장.

Step 3 · Rank the surprise.

강사: *"Which defense surprised you the most? Why is it more clever than the others?"*

코멘트. 가장 놀라운 방어는? 왜 더 영리한가?

STUDENT PROMPTS

- "Why don't plants just run away from danger?"
  식물은 왜 위험에서 도망치지 않는가?
- "Which defense doesn't HURT the animal at all?"
  동물을 전혀 다치게 하지 않는 방어는?
- "Is the acacia tree alone — or working with ants? Why does that matter?"
  아카시아는 혼자인가, 개미와 함께인가? 왜 중요한가?

DIFFERENTIATION

STRUGGLING Pre-fill 3 plant pictures (rose, poison ivy, sensitive plant). Match each to "sharp / chemical / helper" tag.

3가지 그림 + 3가지 분류 매칭.

ADVANCED Invent a new defense a plant could have. Name it and explain HOW it would stop animals — write 3 sentences.

새로운 방어 발명 — 이름·작동 방식 3문장.

COMMON MISCONCEPTIONS

PITFALL "Plants are completely helpless because they can't move." Many plants have surprising defenses — sharp, chemical, or helper-based.

식물은 무력하지 않음. 날카로움·화학·도우미 — 다양한 방어가 있음.

PITFALL "Spines are the only defense." Spines/thorns/prickles are ONE kind. Chemicals and ants are others.

가시만이 방어가 아님. 화학·개미 동맹도 있음.

LEARNER PROFILE LINK

THINKERS Categorizing defenses (sharp / chemical / helper) is critical thinking applied to nature.

방어를 3가지 종류로 분류하는 사고 — Thinkers.

KOREAN BRIDGE

표현 "The plant can't run, but it can protect itself." → "식물은 도망칠 수 없지만, 스스로를 지킬 수 있어."

G2 학생도 이해 가능한 핵심 문장 — '수동/능동'의 새로운 시각.

Q5 Reflection DOK 4

Essential Question: What do plants need to live and grow? Use ideas from TWO of the texts in this module.

MODEL ANSWER

Plants need TWO kinds of things to live and grow. They need the FOUR basic needs — air, water, sunlight, and minerals. AND they need ways to PROTECT those needs from the world around them. In **"Experiment with What a Plant Needs to Grow,"** Nadia Higgins shows the four needs with real experiments. When petroleum jelly blocked AIR, the leaves wilted. When water softened a seed coat, the seed could SPROUT. Without one of the four, the plant fails. In **"Don't Touch Me!,"** Elizabeth Preston shows the second kind. A plant can have all four needs and still die — if a hungry animal eats it. So plants grew SHARP prickles, POISONOUS leaves, and even ANT helpers. These defenses protect the plant long enough for it to USE its four needs. **My Big Idea:** A plant lives by what it takes IN (air, water, sun, minerals) — and by what it keeps OUT (animals that would eat it). Living things grow when both halves are working.

KEYWORDS (3)

4 needs IN — air, water, sun, minerals

defenses OUT — sharp, chemical, helper

both halves together = survive

SCORING NOTES

- Full credit (4): two texts named + evidence from each + personal Big Idea + real-world bridge.
- Partial (3): two texts + effect, Big Idea weak.
- Developing (2): one text only.
- Minimal (1): generic ("plants need water") without text evidence.

SYNTHESIS FRAME

TEXT 1 EVIDENCE

One short detail: a seed coat softens, a leaf wilts under petroleum jelly, soil provides minerals.

CONNECTING WORD

AND / BUT / SO — chooses how needs and defenses fit together.

TEXT 2 EVIDENCE

One short detail: rose prickles, holly poison, sensitive folding, acacia + ants.

REAL-WORLD CONNECTION

KOREA Korean kitchen-garden tradition — 옥상 텃밭 / 화분 키우기. Every Korean apartment plant balances watering, sunlight near a window, and pest defense.

한국 옥상 텃밭과 화분 — 햇빛·물·해충 막기 — 식물 4가지 + 방어가 그대로 일치.

PERSONAL Pick ONE plant in your home or school. Name its four needs AND its one biggest threat.

집 또는 학교 식물 한 가지 — 4가지 필요와 가장 큰 위협 한 가지 명명.

RUBRIC

LEVEL	WHAT IT LOOKS LIKE
Exceeding (4)	Names 2 texts + IN-needs AND OUT-defenses + personal Big Idea + Korea/personal link.
Meeting (3)	Names 2 texts + 4 needs OR defenses + Big Idea.
Approaching (2)	Names 2 texts but only basic needs listed; defenses missing.
Beginning (1)	Generic ("plants need water and sun") without text evidence.

Watch out: Many students will name only the four needs. Push them to also name what plants must KEEP OUT — the second informational text ("Don't Touch Me!") makes the EQ richer. The full answer needs BOTH halves.

TEACHING STEPS

Step 1 · Two-column thinking.

강사: *"For each text, fill two columns: what the plant needs IN / what it keeps OUT."*

코멘트. 텍스트별 2열 — 받아들이는 것 / 막아내는 것.

Step 2 · Find the overlap.

강사: *"Circle ONE thing both texts agree on. (Hint: sunlight, water, or protection.)"*

코멘트. 두 텍스트가 동의하는 한 가지 — 햇빛·물·보호 중에서.

Step 3 · One Big Idea.

강사: *"Write your Big Idea — one sentence. Use IN and OUT."*

코멘트. 한 문장 Big Idea — IN과 OUT 모두 사용.

STUDENT PROMPTS

- "Can a plant survive with all four needs but no defenses?"
  4가지 필요는 있지만 방어가 없는 식물은 살 수 있을까?
- "Can a plant survive with defenses but no sunlight?"
  방어는 있지만 햇빛이 없는 식물은 살 수 있을까?
- "What is one thing a plant needs that you also need?"
  식물이 필요로 하는 것 중 나도 필요한 것 하나?

DIFFERENTIATION

STRUGGLING Sentence frame: "In ___, the plant needs ___. In ___, the plant keeps out ___. So a plant lives by ___ AND ___."
5단 문장 골격.

ADVANCED Compare the FOREST passage (D-Passage 4) with the EXPERIMENT text. Both show photosynthesis at different scales — one bean vs. a whole forest. Which view is "more true"? Why?
실험 텍스트 vs. 숲 지문 — 같은 광합성, 다른 스케일. 어느 쪽이 '더 진실'한가?

COMMON MISCONCEPTIONS

PITFALL "Plants are passive — they just sit there." Plants TAKE IN (photosynthesis) AND KEEP OUT (defenses) — both are active.

식물은 수동적이지 않음. 받아들이고(광합성) 막아내고(방어) — 둘 다 능동.

PITFALL "Defenses are extra — not really needed." A plant with no defense gets eaten before it can use its four needs. Defense is part of survival.

방어는 부수적이 아님. 방어 없이 먹히면 4가지 필요도 소용 없음.

LEARNER PROFILE LINK

REFLECTIVE Holding "what comes IN" and "what stays OUT" together is reflective thinking about living systems.

'들어오는 것'과 '나가지 않는 것'을 함께 잡는 사고 — Reflective.

CARING A child who can grow a plant and protect it is caring — for life that cannot speak.

말하지 못하는 생명을 돌보는 마음 — Caring.

KOREAN BRIDGE

표현 "A plant lives by what comes in and what stays out." → "식물은 들어오는 것과 나가지 못하게 막는 것으로 살아."

EQ에 대한 가장 강한 학생 답변 — 글쓰기 마무리.

관용구 "Time to grow!" → "이제 자랄 시간!"

모듈 제목 — 한국어 그대로 활용 가능.

Worksheet D — MAP Practice · Answer Key

20 items · RIT 180 → 200 · Lexile 240L–500L

PASSAGE 1 · THE BEAN IN THE CUP

Q1 RIT 180

A

Lexile 240L · Key Detail. "Her teacher gave her one bean."

Q2 RIT 184

C

Lexile 250L · Cause & Effect. Windowsill = sunlight; sunlight helps a plant grow.

Q3 RIT 186

B

Lexile 260L · Inference. The seed coat was softening under the soil — germination happens unseen.

Q4 RIT 188

D

Lexile 270L · Vocabulary. Section A Q2 word reused — "germinate" = begin to grow.

Q5 RIT 189

C

Lexile 290L · Main Idea. Dirt + water + sun → seedling. Mirrors the EQ.

PASSAGE 2 · THE CACTUS AND THE BOY

Q6 RIT 190

B

Lexile 300L · Key Detail. "Tomas lived in a hot, dry place."

Q7 RIT 192

D

Lexile 320L · Cause & Effect. "The spines keep deer and goats away... The spines protect that water."

Q8 RIT 193

A

Lexile 330L · Vocabulary. Section A Q24 word reused — "poke" = press with a sharp point.

Q9 RIT 194

C

Lexile 340L · Inference. Stored water = survival in a place with little rain.

Q10 RIT 195

B

Lexile 350L · Main Idea. Spines + stored water = both halves of the EQ (defense + need).

PASSAGE 3 · THE TOMATO THAT ALMOST DIED

Q11 RIT 195

D

Lexile 360L · Key Detail. "His grandfather gave him a tomato seedling."

Q12 RIT 196

A

Lexile 370L · Cause & Effect. Missed weekend watering → soil bone-dry → seedling drooped.

Q13 RIT 197

C

Lexile 380L · Vocabulary. Big Idea Word Q1 reused — "fertilize" = add something to soil that helps plants grow.

Q14 RIT 198

B

Lexile 390L · Inference. Water + fertilizer (minerals) restored — both needs returned together.

Q15 RIT 199

A

Lexile 410L · Main Idea. Daily care = the 4 needs every day. Shows-up theme.

PASSAGE 4 · THE FOREST THAT CLEANS THE AIR

Q16 RIT 199

D

Lexile 440L · Key Detail. "The leaves take in a gas called carbon dioxide."

Q17 RIT 200

A

Lexile 460L · Cause & Effect. Trees feed themselves AND produce oxygen as a side effect — like a factory.

Q18 RIT 200

B

Lexile 480L · Vocabulary. Section A Q5 trap word reused — "process" = set of steps in order.

Q19 RIT 200

D

Lexile 490L · Inference. Roots hold soil; leaves slow rain — without forests, soil washes away.

Q20 RIT 200

C

Lexile 500L · Main Idea. Forests = same needs as one plant, but at planet scale — they answer the EQ for the world.